

Methodology of Outside Field Dosimetry

Acuros as algorithm and Halcyon as a LINAC



Souvik Das

Resident Medical Physicist
Tata Memorial Hospital, Mumbai

Out of Field Dosimetry in Halcyon using Acuros as algorithm.

Equipment required:

- Halcyon LINAC
- Eclipse Treatment Planning System
- ATOM anthropomorphic male reference phantom (CIRS, Inc., Norfolk, VA)
- LD

Which Site to be investigated?

I have to make a dataset which type of patient are lived long after the radiation therapy.

ATOM anthropomorphic male reference phantom

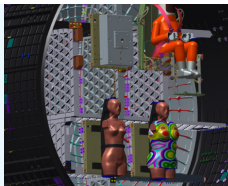
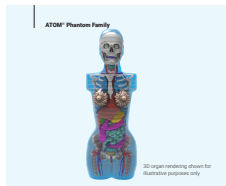
Description	Model	Available Dosimetry Configurations	Height, cm	Weight, kg	Thorax Dimensions, cm	Bone Physical Density, cc	Bone Electron Density, 1/cc
Adult Male	701	B, C, D & G	173	73	23 cm $\times$ 32 cm	1.60	5.030×10^{23}

Configuration key:

- B Configuration = 3 $\times$ 3 cm Grid - 5 mm Holes
- C Configuration = 1.5 $\times$ 1.5 cm Grid - 5 mm Holes
- D Configuration = Organ Grid - 5 mm Holes
- G Configuration = Organ Grid - 14 mm Holes

ATOM anthropomorphic male reference phantom

Manufacturer slide



Enabling Safer Human Space Travel

NASA's Artemis I mission to the Moon in 2022 featured two custom designed female ATOM Phantoms. "Helga" and "Zohar" flew as proxies for female astronauts to help determine risk to radiation sensitive organs, and ensure more inclusive possibilities for the future of human space travel.

Learn more: sunnuclear.com/news

Images source: The European Space Agency and DLR



Plug Configurations

Expanding Functionality

Regardless of the grid configuration or hole diameter, all holes in an ATOM phantom come profiled with an appropriate, tissue equivalent solid plug. These solid plugs can be easily removed and replaced with a dosimeter holder plug for available dosimetry configurations. These special plugs are designed to house a variety of dosimeters, including TLD chips, TLD rods, TLD glass rods, MOSFET detectors, and OSL holders. This pairing of dosimeter and plug protects the dosimeter and allows it to be placed in any hole in the phantom, with minimal air interface between the device and surrounding tissue.

All plugs are sold as kits that contain a variety of tissue types to ensure there is an equivalent plug for any dosimetry configuration and anatomical location

Thermoluminescence Dosimetry (TLD)

Sectional and drilled ATOM phantoms readily accept TLDs using a 5 mm plug. These plugs are manufactured to hold TLD chips, rods, bars, cubes, and discs in place during the dosimetry study. In applications where varying the depth of the dosimeter within the slab is important, such as in organ dosimetry, these rods can be easily modified by the user to custom depths within the slab.



Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

This specially machined plug can hold MOSFET dosimeters. The plug is manufactured to fit 5 mm diameter holes in an ATOM phantom. A recessed section at the center of the plug accepts the MOSFET sensitive volume and a radius on the end of the plug prevents harsh bends in the cable to prevent damage to the cable when the phantom is assembled.

OSL Dosimetry

OSL holder plugs are designed to fit standard OSL dosimeters and require 14 mm diameter hole configurations to accommodate the device.

Ion Chamber Dosimetry

ATOM phantoms can be custom drilled to accommodate various ion chambers and diodes. The Chamber cavity is drilled in the lateral direction along the midplane of the section. Each chamber cavity is supplied with a soft tissue equivalent plug. Upon order, customers should specify the desired anatomical location of the sensitive volume for the ion chamber as the location of the chamber will affect some holes in the grid configuration.

Description	Part #	Plug Size	Kit Contents
TLD Chip		5 mm Ø x 25 mm long	
TLD / Glass Rod		5 mm Ø x 25 mm long (Ø 1.5mm Ø x 12 mm deep)	
MOSFET		5 mm Ø x 25 mm long	5 sets of 10 plugs (soft tissue, lung, brain, adult bone and pediatric bone plugs)
OSL		14 mm Ø x 25 mm long	5 sets of 10 plugs (soft tissue, lung, brain, adult bone and pediatric bone plugs)
Ion Chambers			

By quotation only, contact customer service for a quote

Which Luminescence Dosimeter(LD)?

See

TABLE I. Thermoluminescent dosimeter (TLD) and optically stimulated luminescent dosimeter (OSLD) materials available commercially along with example commercial names, density ρ , effective atomic number Z_{eff} , temperature of the main TL peak, and typical emission wavelength, sensitivity relative to LiF, beam quality dependence (k_Q ; response at 30 keV vs that in cobalt), and fading. Tabulated values based on data from the literature.^{2,28-31}

	Commercial names ^a	ρ (g/cm ³)	Z_{eff} ^b	Main glow peak temp ^c (°C)	Emission ^d (nm)	TL sensitivity vs. LiF	k_Q 30 keV/ ⁶⁰ Co	Fading of dosimetry peak at normal temp
LiF:Mg,Ti (TL)	TLD-100	2.6	8.31	~235	~410	Referent	1.3	5% in 3–12 months
⁶ LiF:Mg,Ti (TL)	TLD-600	2.6	8.31	~235	~410	1.0	1.3	5% in 3–12 months
⁷ LiF:Mg,Ti (TL)	TLD-700	2.6	8.31	~235	~410	1.0	1.3	5% in 3–12 months
LiF:Mg,Cu,P (TL)	TLD-100H	2.5	8.31	~200	~370	30	1.25	2% in 3 months
Li ₂ B ₄ O ₇ :Mn (TL)	TLD-800	2.3	7.32	~185	~600	0.3	0.9	5–10% in 3 months
CaF ₂ :Dy (TL)	TLD-200	3.18	16.90	~160, 185, 245, 290	480, 575, 660, 750	30	14	25% in 4 week
CaF ₂ :Mn (TL)	TLD-400	3.18	16.90	~300	~495	10	14	15% in 2–4 week
CaSO ₄ :Dy (TL)	TLD-900	2.61	15.62	~220	480, 575, 660, 750	15	12	6% in 6 months
Al ₂ O ₃ :C (OSL)	nanoDot™	3.95 ^e	11.28	~200	~410	N/A	3.7	4% in 3 months
BeO (OSL)	Thermalox 995	2.85	7.21	~210, 330	~335 (TL), ~390 (OSL)	N/A	0.82	5–10% in 3 months

^aCommercial name used by Thermo Fisher Scientific, Inc. (formerly Harshaw, Inc.), Materion Ceramics, and Landauer Inc.

^bValues from Bos.²

^cThese are approximate values, as the TL peak temperature varies with heating rate and may also vary considerably with impurity content and source of material.

^dThe symbol “~” indicates broad emission bands. Emission from Dy³⁺ in Dy-doped materials are characterized by sharp emission lines.

^eThe density of Al₂O₃ is 3.95; however, the active nanoDot crystal is powder embedded in a polyester tape and includes many air pockets.³² The density of the active volume has been found to be in the range of 1.41–2.45 g/cm.^{33,34}

Figure: [1]

Dose at the surface

- Outside the treatment field, the dose at the surface is increased by **stray electrons**, so there is a build-down effect instead of a build-up effect near the surface.
- The dose is elevated by a factor of as much as 5 at the patient surface, and decreases to a depth of approximately d_{max} , below which the dose becomes approximately constant with depth
- Therefore, if a dosimeter is placed on the patient surface, it will overestimate the dose (by a factor of as much as 5) unless a very superficial dose estimate is desired. If a superficial dose estimate is not desired, the dosimeter should be covered by bolus of a thickness of approximately d_{max} (For halcyon $d_{max} = 1.3$ cm). [2]

- The average energy is much lower outside the treatment field (mean photon energy as low as 0.20 MeV).
- If the dosimeter's calibration is done in the harder, primary beam (mean photon energy 1.5 MeV for a 6 MV beam), a dosimeter with a higher effective atomic number than tissue will overrespond to this softer radiation.
- While this means that measurements will tend to be conservative overestimates, this effect must at least be known, and can be sizeable to the point of unacceptable accuracy unless it is accounted for.

At 6 MV, TLD-100 over-responds by 4-12% increasing with greater distances from the field edge.

Variations in photon energy spectra of a 6 MV beam and their impact on TLD response

Sarah B. Scarboro, David S. Followill, Rebecca M. Howell, and Stephen F. Kry^{a)}

The University of Texas MD Anderson Cancer Center, Houston, Texas 77030 and The University of Texas Health Science Center Houston, Graduate School of Biomedical Sciences, Houston, Texas 77030

(Received 28 October 2010; revised 18 March 2011; accepted for publication 21 March 2011; published 6 May 2011)

Purpose: Measurement of the absorbed dose from radiotherapy beams is an essential component of providing safe and reproducible treatment. For an energy-dependent dosimeter such as thermoluminescent dosimeters (TLDs), it is generally assumed that the energy spectrum is constant throughout the treatment field and is unperturbed by field size, depth, field modulation, or heterogeneities. However, this does not reflect reality and introduces error into clinical dose measurements. The purpose of this study was to evaluate the variability in the energy spectrum of a Varian 6 MV beam and to evaluate the impact of these variations in photon energy spectra on the response of a common energy-dependent dosimeter, TLD.

Methods: Using Monte Carlo methods, we calculated variations in the photon energy spectra of a 6 MV beam as a result of variations of treatment parameters, including field size, measurement location, the presence of heterogeneities, and field modulation. The impact of these spectral variations on the response of the TLD is largely based on increased photoelectric effect in the dosimeter, and this impact was calculated using Burlin cavity theory. Measurements of the energy response were also made to determine the additional energy response due to all intrinsic and secondary effects.

Results: For most in-field measurements, regardless of treatment parameter, the dosimeter response was not significantly affected by the spectral variations (<1% effect). **For measurement points outside of the treatment field, where the spectrum is softer, the TLD over-responded by up to 12% due to an increased probability of photoelectric effect in the TLD material as well as inherent ionization density effects that play a role at low photon energies.**

Conclusions: It is generally acceptable to ignore the impact of variations in the photon spectrum on the measured dose for locations within the treatment field. However, outside the treatment field, the spectra are much softer, and a correction factor is generally appropriate. The results of this work have determined values for this factor, which range from 0.88 to 0.99 depending on the specific irradiation conditions. © 2011 American Association of Physicists in Medicine.

[DOI: 10.1118/1.3575419]

Figure: [3]

Out-of-field photon dose following removal of the flattening filter from a medical accelerator

Stephen F Kry¹, Oleg N Vassiliev and Radhe Mohan

Department of Radiation Physics, The University of Texas MD Anderson Cancer Center, Houston, TX, USA

E-mail: skry@mdanderson.org

Received 3 December 2009, in final form 11 February 2010

Published 19 March 2010

Online at stacks.iop.org/PMB/55/2155

Abstract

The aim of this paper is to determine the effect of removing the flattening filter from a linear accelerator on the out-of-field photon dose. A Monte Carlo model was used to simulate 6 MV and 18 MV photon beams from a Varian 2100 accelerator with the flattening filter in place and with it removed. The out-of-field photon doses and composition (head leakage, patient scatter and collimator scatter) were calculated from square open fields in a water tank as a function of distance from central axis, field size and depth. The out-of-field doses resulting from intensity-modulated radiation therapy to the prostate at 6 MV were also calculated, with and without the flattening filter, to sensitive organs in an anthropomorphic Rando phantom. Removal of the flattening filter reduced the out-of-field dose near the treatment field (<3 cm from the field edge) because of decreased collimator scatter. It increased the out-of-field dose at intermediate distances from the field edge (3–15 cm) because of increased patient scatter. At greater distances, the out-of-field dose was decreased because of reduced head leakage. For the clinical treatment examined, the out-of-field dose was generally reduced following removal of the flattening filter, particularly at large distances from the treatment field. Removal of the flattening filter may be advantageous by reducing the out-of-field dose to the patient. This could contribute to reducing the long-term risk of secondary malignancies. In general, however, the out-of-field dose depends on treatment and patient parameters, and a reduction may not always be achievable.

Figure: [4]

Howell's study [5]

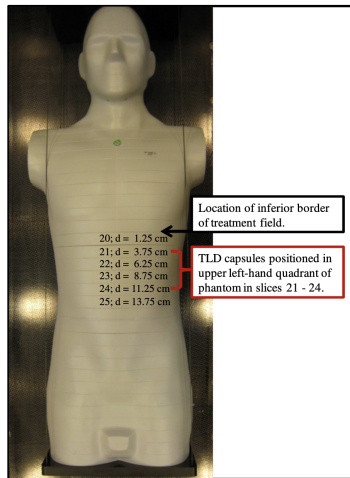


Figure: The inferior edge of the mantle fields abutted the superior border of slice 20 of the phantom. [5]

Howell's study

continued

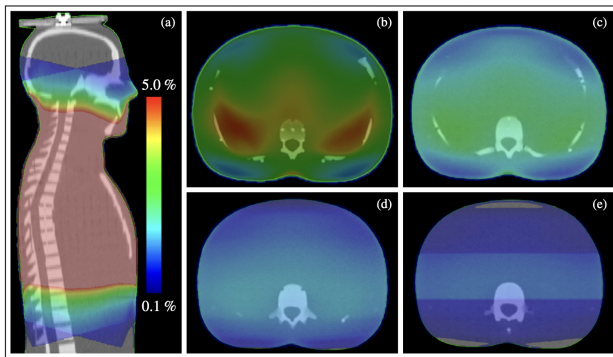


Figure 2. Out-of-field dose reported by the treatment planning system. Dose scale is set to show 5% to 0.1% of the prescription dose. Dose color wash distributions are shown for (a) sagittal view and axial plane at the center of phantom slices: (b) slice 21, (c) slice 22, (d) slice 23 and (e) slice 24. Note that all dose values above 5% are at the center of the phantom and are shown in dark pink.

Howell's study

continued

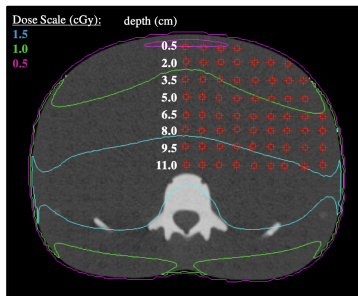


Figure 3. Axial image for slice 23 of the phantom corresponding to the center of the phantom, with the data point grid shown by red squares. Within the TPS, point doses were determined at each location on the grid using the point dose measurement tool. Each point within the point dose grid corresponds to a single TLD capsule lying at the equivalent position in the phantom.

Figure: A total of 30 Gy was delivered to the treatment isocenter in a single irradiation using 6 MV photons. This corresponded to a range of doses to the **LiF TLD-100** of 50-500 cGy.

Howell's study

continued

Table 1. Mean measured doses ($\mu_{\text{meas}} \pm \sigma$) and mean TPS-calculated doses ($\mu_{\text{calc}} \pm \sigma$) for all TLD data for each phantom slice and for all phantom slices. The standard deviation (σ) is reported as one standard deviation of the mean. **This value is dominated by the spread of doses across each phantom slice as compared to an additional standard uncertainty in each TLD measurement of $\leq 3\%$.**

Phantom slice	Distance from field edge (cm)	Count	$\mu_{\text{calc}}, \text{cGy/Gy}_{\text{Rx}} (\sigma)$		$\mu_{\text{meas}}, \text{cGy/Gy}_{\text{RX}} (\sigma)$		Mean TPS underestimation of measured dose (σ)	
21	3.75	56	3.08	(0.61)	4.24	(0.45)	28%	(17%)
22	6.25	59	2.02	(0.43)	3.01	(0.24)	32%	(12%)
23	8.75	62	1.16	(0.32)	2.09	(0.14)	44%	(15%)
24	11.25	61	0.66	(0.33)	1.49	(0.13)	55%	(23%)
All slices	N/A	238	1.7	(1.01)	2.65	(1.05)	40%	(20%)

cGy/Gy_{Rx}: cGy per prescribed Gy.

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